

Fashion-MNIST Image Classification Using ANN

1. Introduction

An artificial neural network classifies Fashion-MNIST images into 10 clothing classes.

2. Dataset and Preprocessing

Fashion-MNIST contains 28 x 28 grayscale images. Pixel values are normalized from 0-255 to 0-1. The Flatten layer converts every image to 784 input features.

3. ANN Architecture

Flatten -> Dense(128, ReLU) -> Dense(64, ReLU) -> Dropout(0.2) -> Dense(10, Softmax)

Optimizer: Adam | Loss: Sparse Categorical Crossentropy | Metric: Accuracy

4. Training Results and Model Evaluation

Modified-model test accuracy: 87.08%

Modified-model test loss: 0.3612

See accuracy.png, loss.png, confusion_matrix.png, classification_report.txt, and prediction_examples.png.

5. Experiment and Comparison

Original model (no dropout) test accuracy: 86.35%

Modified model (Dropout 0.2) test accuracy: 87.08%

The experiment changes dropout only; compare the two test accuracies to judge its effect.

6. Streamlit Application

The app loads the saved Keras model, preprocesses an uploaded image, and shows the predicted class and confidence.

7. Conclusion

The ANN provides a complete Fashion-MNIST classification workflow and an interactive prediction interface.